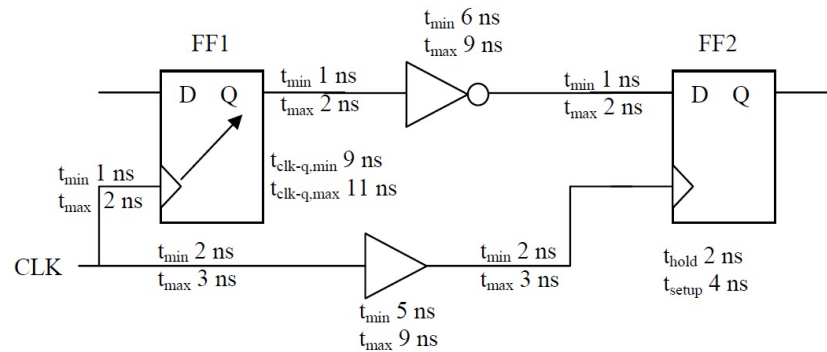
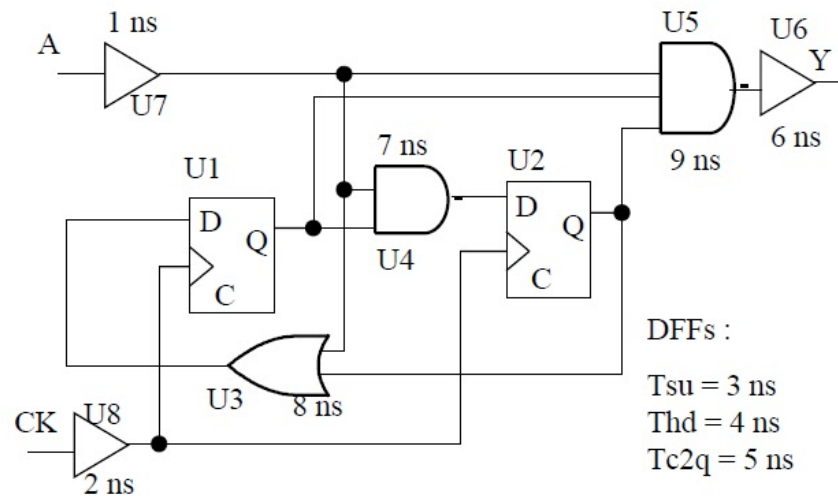


Question 1

In the following circuit, find out whether there is any setup or hold violations. Consider T_{CLK} as 15 ns and 22 ns.

**Question 2**

In order to work correctly, what should be the setup and hold time at input A in the following circuit? Also find out maximum operating frequency of this circuit.

**Question 3**

In the above circuit, try to improve the timing by adding any buffer or register.